



# FindHUB

## AI Powered Campus Lost & Found

~ Because every lost item deserves to be found.

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# How Broken Is the Current System?

## Manual Reporting

Items are logged by hand with no standardization, leading to inconsistent records.

## No Smart Matching

Lost and found items sit in separate silos with no automated way to connect them.

## Time-Consuming

Users must physically visit locations, call offices, or browse paper logs.

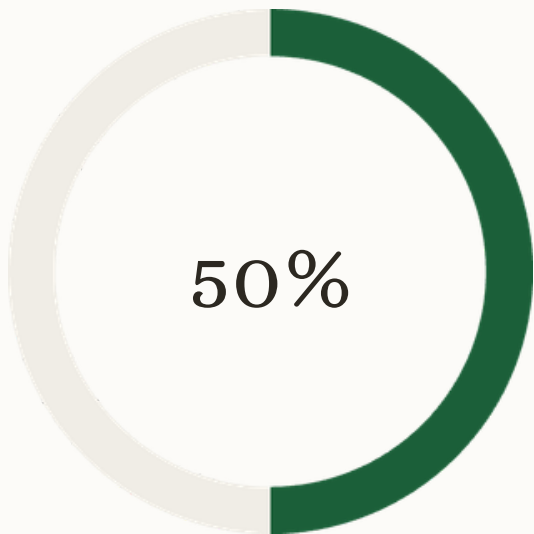
## No Digital Bridge

There's no platform to instantly notify potential matches between users.

# Lost Items $\neq$ Found Items

# The Problem Is Real

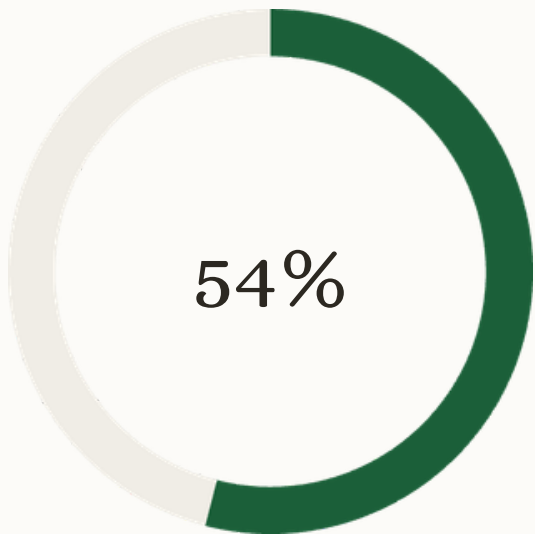
Campus theft and lost items represent a **pervasive, unsolved problem** affecting millions of students nationwide. Despite being the #1 campus crime, no intelligent, centralized system exists to effectively track and recover lost property.



~By Reports from  
National Center for  
Education Statistics  
(NCES)

Students Affected

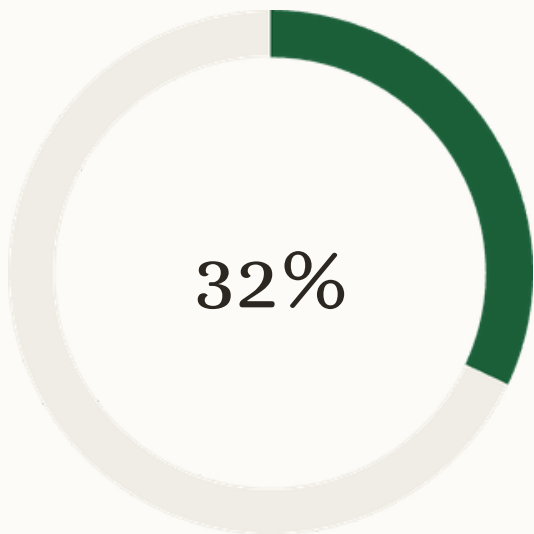
Lose something on campus every academic year



~By Reports from  
Utah Valley University  
campus survey data  
(UVU)

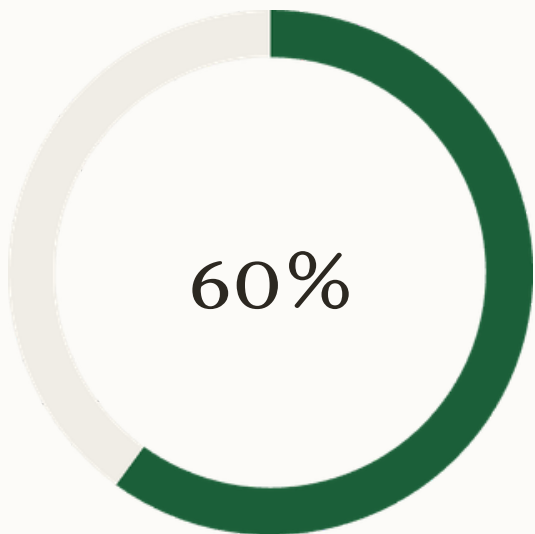
Have Items Stolen

Report theft from campus locations (UVU Study)



Returned Within 24h

Lost items are recovered — the rest go unclaimed



Give Up Searching

Don't even attempt to recover lost items

**Theft accounts for 1 in 4 reported campus incidents, yet students continue to search through scattered bulletin boards and fragmented social media groups with no coordinated solution.**

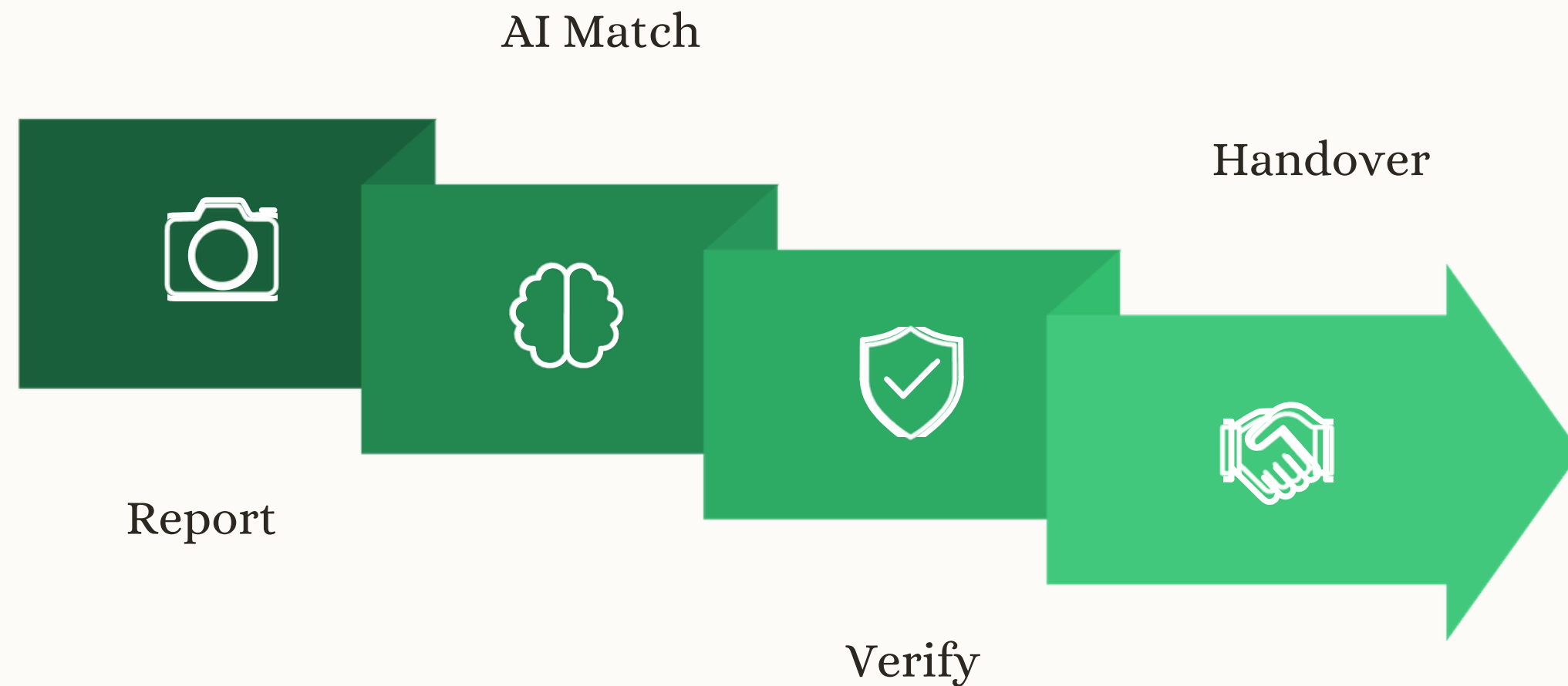
# Lost it? FindHUB it.

The smart campus lost and found. We use AI to confidentially match lost items to found items, keeping thieves out and your stuff safe.

- CLIP - Based Image Matching
- NLP-enhanced text matching
- AI anti-theft verification
- Hidden found-item inventory
- Google Maps location pinning
- Community threads
- AI insight extraction
- Notifications + case status tracking

# Introducing FindHUB

An AI-powered platform that **automatically matches lost and found items** using vision and language models, verifies true ownership through intelligent questioning, and enables seamless handover — all without exposing items to public theft risk.



**From report to reunion — fully automated, fraud-resistant, and campus-ready.**



# Key Features at a Glance



## CLIP Image Matching

Vectorizes images with cosine similarity — **~80% top-1 retrieval accuracy**.



## AI Ownership Verification

LLM generates secret questions from hidden clues — only the true owner can answer.



## Real-Time Chat

Secure Socket.io channels open post-verification for handover coordination.



## NLP Text Analysis

spaCy + regex parses descriptions for multi-criteria scoring.



## Google Maps Integration

Pin exact report locations on an interactive campus map.

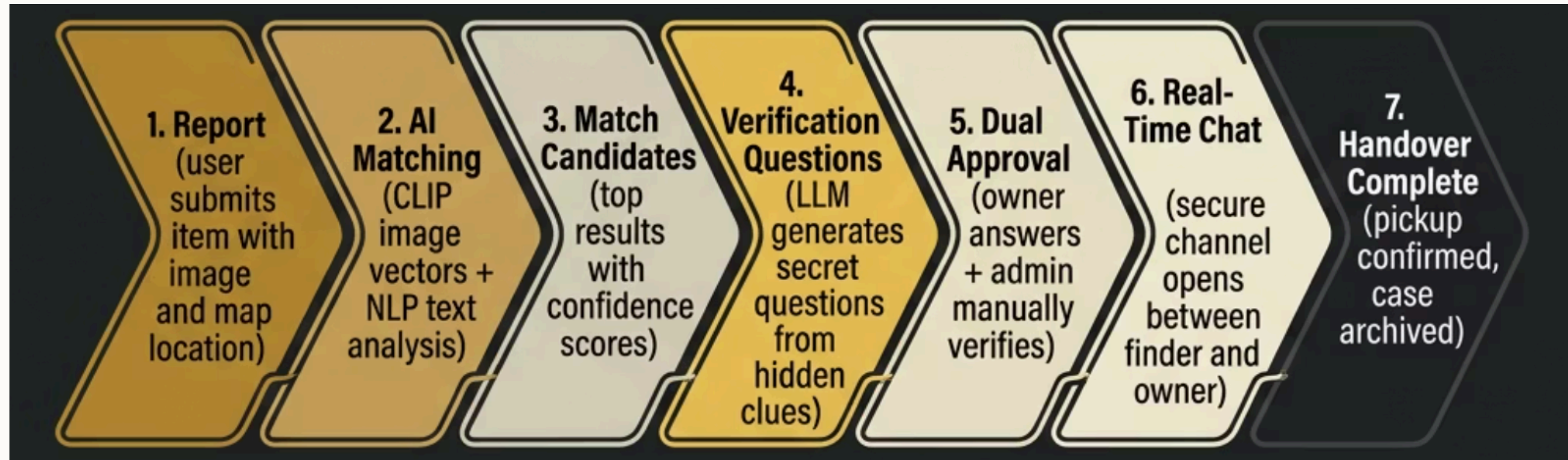


## Community Thread AI

AI analyzes full conversations, extracting sightings, patterns, and behavioral clues.



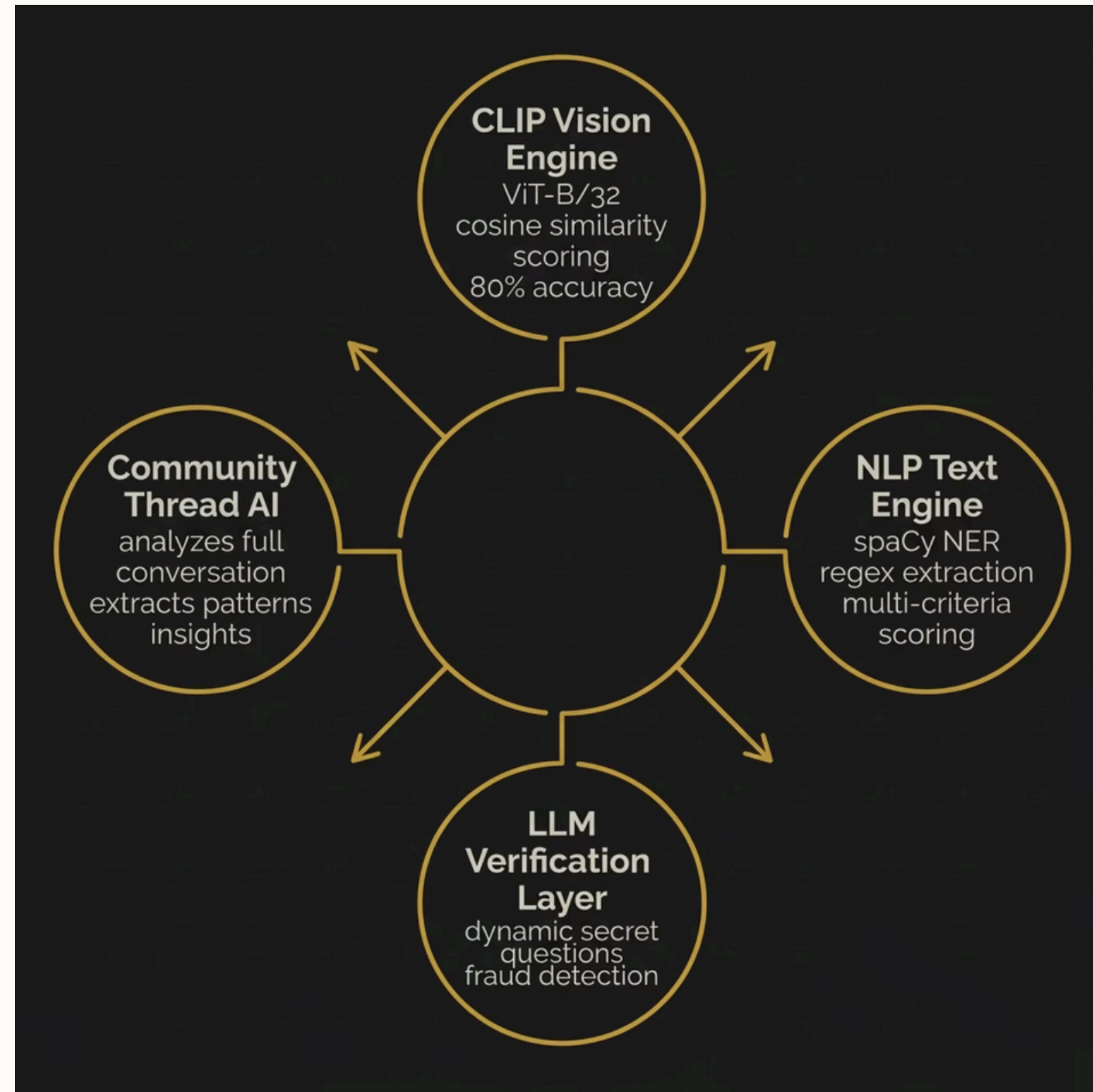
# How It Works: The Core Pipeline



Every step is designed to maximize match accuracy while eliminating fraud. The pipeline runs automatically on every new submission — and re-matches periodically as the database grows.

**Hidden Found Items:** Found items are **never publicly visible**. Only admins see the full list — preventing opportunistic theft before verification even begins.

# AI & ML: Under the Hood



## Why This Stack Works

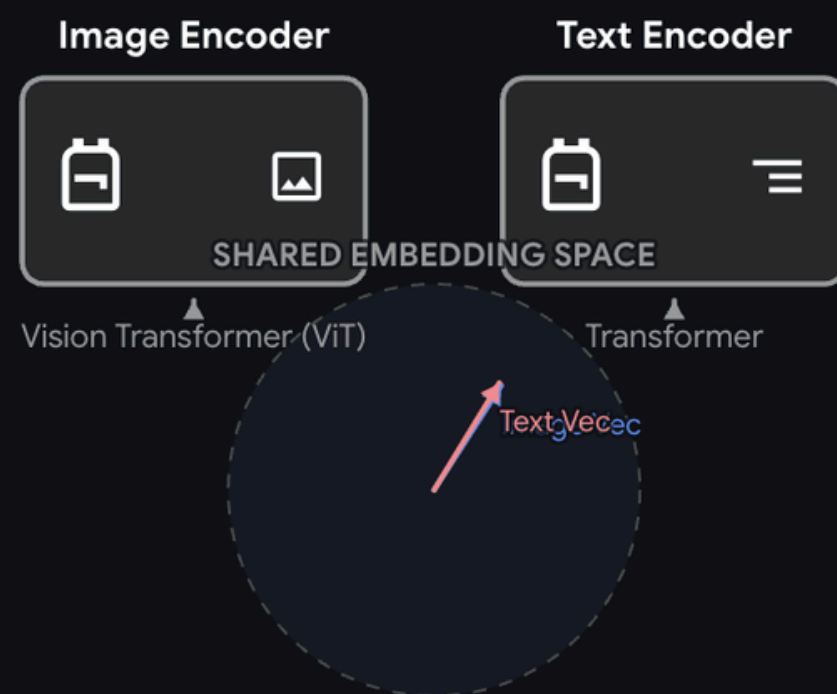
- **CLIP ViT-B/32** : production-grade visual embeddings, ~80% top-1 retrieval accuracy on benchmarks
- **spaCy + regex** : structured entity extraction from natural language descriptions
- **LLM question generation** : dynamic, clue-based verification questions no fraudster can anticipate
- **Thread intelligence** : full-conversation AI analysis, not just location, extracts sightings, patterns, and behavioral clues
- **Periodic re-matching** : engine re-runs as new items enter the system, catching delayed matches



# CLIP: Contrastive Language-Image Pre-training

SIMILARITY STATUS  
0.88 Match Found

Match Found



CONTRASTIVE BATCH MATRIX

Similarity =  $(A \cdot B) / (\|A\| \|B\|)$

|          |      |      |      |
|----------|------|------|------|
| Water    | 0.18 | 0.88 | 0.25 |
| Backpack | 0.09 | 0.15 | 0.95 |
| Keys     | 0.09 | 0.15 | 0.95 |

hydroflaskred of

Image Input

Red Backpack

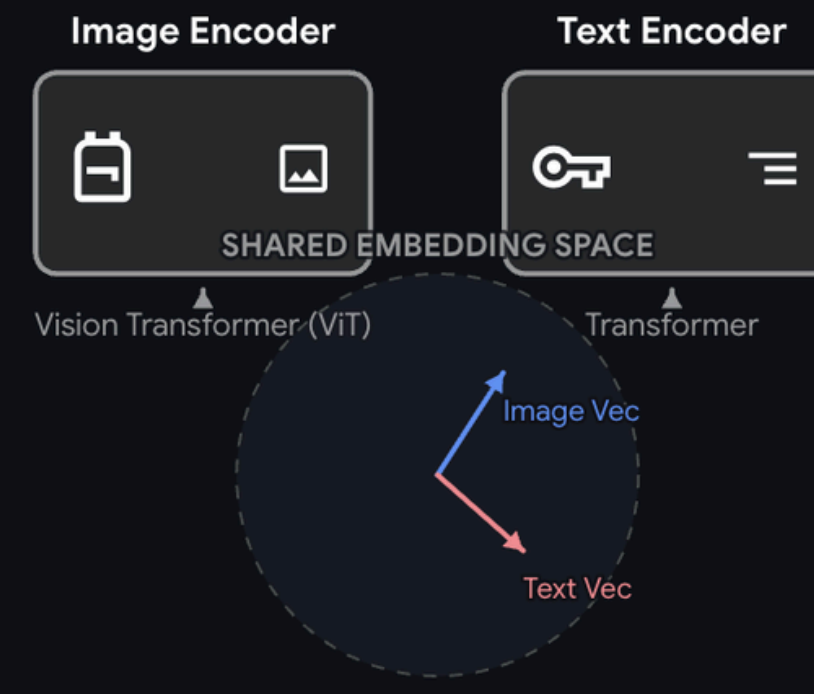
Text Input

A bright red school bag

# CLIP: Contrastive Language-Image Pre-training

SIMILARITY STATUS  
0.25 No Match

No Match



CONTRASTIVE BATCH MATRIX

Similarity =  $(A \cdot B) / (\|A\| \|B\|)$

|          |      |      |      |
|----------|------|------|------|
| Water    | 0.92 | 0.18 | 0.25 |
| Backpack | 0.18 | 0.88 | 0.25 |
| Keys     | 0.09 | 0.15 | 0.95 |

hydroflaskred of

Image Input

Red Backpack

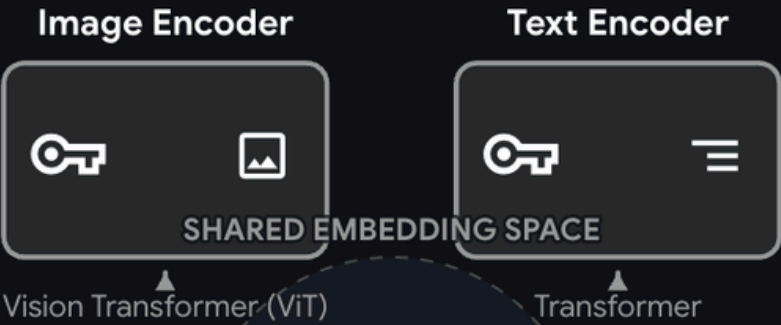
Text Input

A set of house keys

CLIP: Contrastive Language-Image Pre-training

SIMILARITY STATUS  
0.95 Match Found

Match Found



CONTRASTIVE BATCH MATRIX  
 $\text{Similarity} = (A \cdot B) / (\|A\| \|B\|)$

|          |      |      |      |
|----------|------|------|------|
| Water    | 0.92 | 0.21 | 0.12 |
| Backpack | 0.18 | 0.88 | 0.25 |
| Keys     | 0.09 | 0.15 | 0.95 |

hydroflaskred of

Image Input

Metal Keys

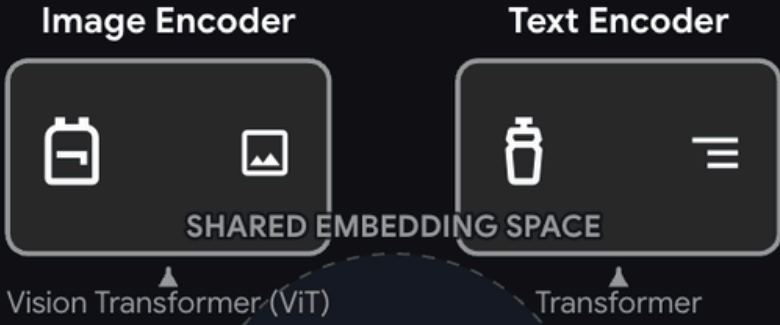
Text Input

A set of house keys

CLIP: Contrastive Language-Image Pre-training

SIMILARITY STATUS  
0.18 No Match

No Match



CONTRASTIVE BATCH MATRIX  
 $\text{Similarity} = (A \cdot B) / (\|A\| \|B\|)$

|          |      |      |      |
|----------|------|------|------|
| Water    | 0.92 | 0.21 | 0.12 |
| Backpack | 0.18 | 0.88 | 0.25 |
| Keys     | 0.09 | 0.15 | 0.95 |

hydroflaskred of

Image Input

Red Backpack

Text Input

A dark hydroflask

## Report Found Item.

Found something on campus? Submit it here. You will not be able to see this report again – it goes into the private found database for matching.

### Item Details

Category \*

Select category...

Item Name \*

e.g. iPhone 15

Brand

e.g. Apple

Color

e.g. Space Gray

Description

Physical condition, packaging, any visible details...

### Secret Clues

Add details only the true owner would know. These are used to generate verification questions.

Clue 1: e.g. "Red sticker on the back"

+ Add another clue

# Product Walkthrough

A guided tour of **FindHUB** in action – from first report to final handover.

### Where and When

Found at \*

e.g. Canteen counter

Tag my GPS location

Date found \*

dd/mm/yyyy

### Images



Click to upload images

Max 3 images, 5MB each · JPEG, PNG, WebP



Stay anonymous

Your identity will be hidden from the owner. Handover via admin.



Submit Found Report

 Lost (1)

 Found (0)



**TShirt**

Clothing · Room 204, Bulding Academic · 7 Apr 2026

MATCHED



1 unread

✓✓ Mark all read



**Potential match found! •**

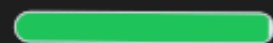
We found a 98% match for your TShirt. Verify ownership now.

5m ago



**TShirt**

Clothing · just now

 98%

 AWAITING VERIFICATION

Verify now →

Verifying match for: TShirt

Attempt 1 of 2



Enter your password to verify your identity. The lock icon next to your password is active.

**Q1: What is the brand of the item?**

Your answer...

**Q2: Does the item have any unique features or marks?**

Your answer...



Submit Verification



**Not Verified**

You have 1 attempt(s) remaining. Try again with more accurate answers.

Try Again

Back to matches





# Anti-Theft by Design

Every layer of **FindHUB** is engineered to prevent false claims and protect genuine owners.

## Hidden Found Items

Found items are **never publicly visible**. Only admins see the full list — eliminating browsing by would-be thieves.

## Secret Clue Verification

Finders add hidden clues (stickers, scratches). The LLM converts these into questions only the real owner can answer.

## Fraud Detection

Repeated false claims trigger automated flags. Sentence similarity scoring catches inconsistent answers.

## Dual Verification + Audit Trails

Both user self-verification and admin manual review are supported. Every action is logged for full accountability.

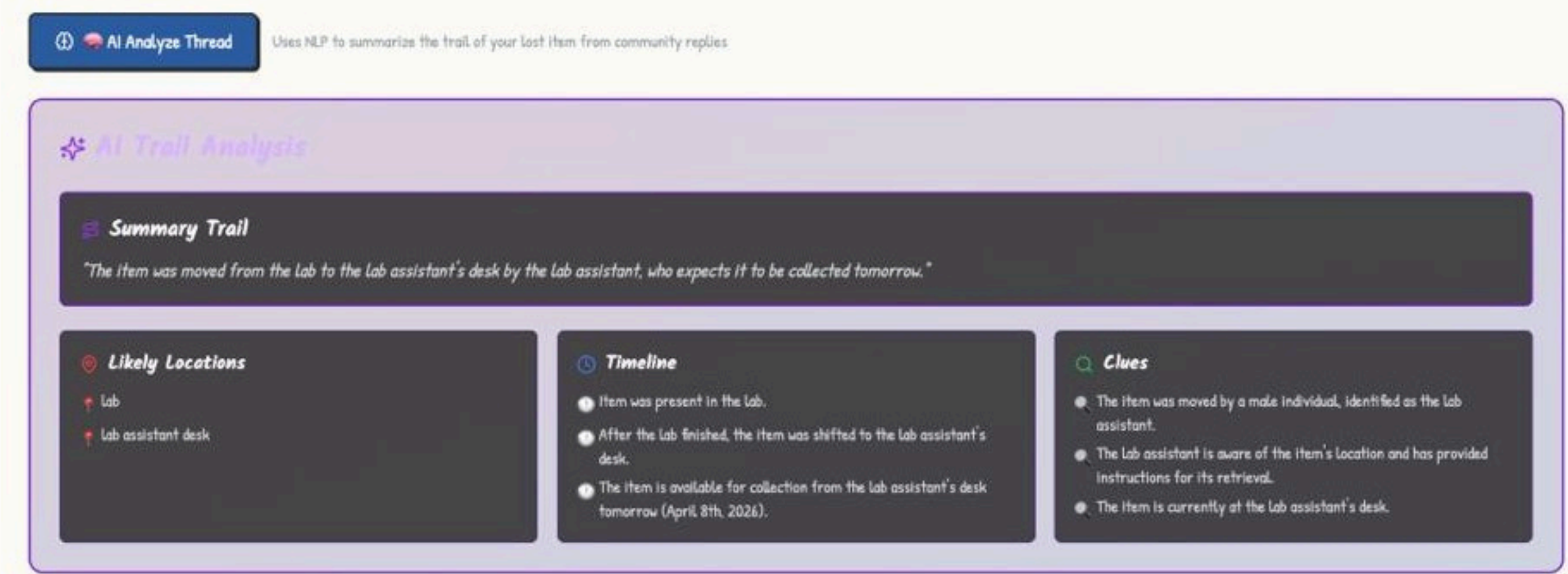




# Community-Powered Matching: AI-Enhanced Crowd Collaboration

Turn community conversations into actionable recovery insights with intelligent thread analysis.

# Full-Thread AI Analysis: Context Matters



## Conversation Context

AI analyzes the entire thread history, understanding how messages relate to each other rather than treating each post in isolation.

## Location Mapping

Geographic references are extracted and mapped to show movement patterns and high-probability search areas.

## Temporal Patterns

The system identifies when sightings cluster in time, revealing active periods when items are more likely to be found.

## Entity Recognition

Key identifiers like brands, colors, and unique features are cataloged to build a comprehensive profile.

# Tech Stack

## Frontend

React + Tailwind CSS

## Backend

Node.js + Express, JWT

## Database

MongoDB

## ML Service

Python + FastAPI (CLIP, spaCy, LLM)

## Maps

Google Maps API

## Cloud & Infra

Clouinary

# Impact, Roadmap & Thank You

## Who Benefits

- **Students** — fast, fair item recovery without theft risk
- **Staff & Admins** — centralized oversight, analytics, and audit trails
- **Campuses** — scalable to multi-campus deployment, integrable with campus security and ID systems

## What's Next

- Mobile app (iOS + Android)
- Campus ID system integration
- Expansion beyond campus settings

## Challenges & Learnings

- **Matching accuracy** — tuned CLIP thresholds and multi-criteria scoring to reduce false positives
- **Real-time performance** — optimized embedding caching for sub-second match results
- **LLM question quality** — iterated on prompt engineering to generate unambiguous verification questions

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## Thank You

Questions? We'd love to walk through the architecture or demo in more detail.